

Chapter 1

Overview of Electronic Business and Electronic Commerce

Learning Objectives

1. Define electronic Business (EB) and describe its various categories.
2. Define electronic commerce (EC) and describe its various categories.
3. Describe and discuss the content and framework of EC.
4. Describe the major types of EC transactions.
5. Discuss e-commerce 2.0.
6. Describe social commerce and social software.

Learning Objectives

6. Understand the elements of the digital world.
7. Describe the drivers of EC as they relate to business pressures and organizational responses.
8. Describe some EC business models.
9. Describe the benefits of EC to organizations, consumers, and society.
10. List and describe the major limitations of EC.

Definition: Electronic Business

(U.S. Census Bureau)

- An **process** that an organization conducts over a **computer-mediated network**
 - **Production** – procurement, ordering, stock replenishment, payment processing, production control, etc.
 - **Customer-focused** – marketing, selling, customer order processing, etc.
 - **Internal or management-focused** – employee service, training, recruiting, information sharing, etc.

Electronic Commerce: Definitions and Concepts

- **electronic commerce (EC)**

The process of buying, selling, or exchanging products, services, or information via computer

- **e-business**

A broader definition of EC that includes not just the buying and selling of goods and services, but also servicing customers, collaborating with business partners, and conducting electronic transactions within an organization

Definition: Electronic Commerce

- Any **transaction** completed over a computer-mediated network that involves the **transfer of ownership or rights to use goods or services**.
- Completed transactions may have a zero price.

Definition: Electronic-Business Infrastructure

- The share of total **economic infrastructure** used to support e-business processes and conduct e-commerce transactions.
 - Hardware
 - Software
 - Telecommunication networks
 - Support services
 - Human resources

New Business Rules from E-Business

- Customers determine everything.
- Knowledge and information become more valuable than physical assets.
- People want more choices of products and services.
- Time is present time, distance is zero.
- Technology and network determine how the business is conducted.

Potential Benefits of E-Business

- More product/service selections
- Higher Productivity
- Improved accessibility and convenience
- Better availability of information
- Shorter lead time
- Improved communication
- Stronger competitive position

Examples of E-Business Applications?



Categories of E- Business Applications

(BW, Nov. 24, 2003)

- Collaboration
- Customer Service
- Customization
- Streamlining
- Management
- Cutting Edge

Examples of E-Business Applications

- Cisco (financial management)
- Nike (product design)
- Jet Blue Airlines (customer service & ticketing)
- Dell Computer (customer ordering & service)
- Boeing (flight manuals, maintenance documents, spare parts to order)
- Garden.com (drop ship)
- AlliedSignal (operations scheduling & supply chain)
- Starbucks (smart cards & web access)

Launching E-Business Initiatives

- Define what an e-business initiative means for the organization.
- Ask key questions to develop a strategy.
- Communicate the plan to executives and employees extensively.
- Prepare for mistakes and develop effective recovery systems.

Electronic Commerce: Definitions and Concepts

- **MAJOR EC CONCEPTS**

- **Pure Versus Partial EC**

- **EC Organizations**

- **brick-and-mortar (old economy) organizations**

- Old-economy organizations (corporations) that perform their primary business offline, selling physical products by means of physical agents

- **virtual (pure-play) organizations**

- Organizations that conduct their business activities solely online

- **click-and-mortar (click-and-brick) organizations**

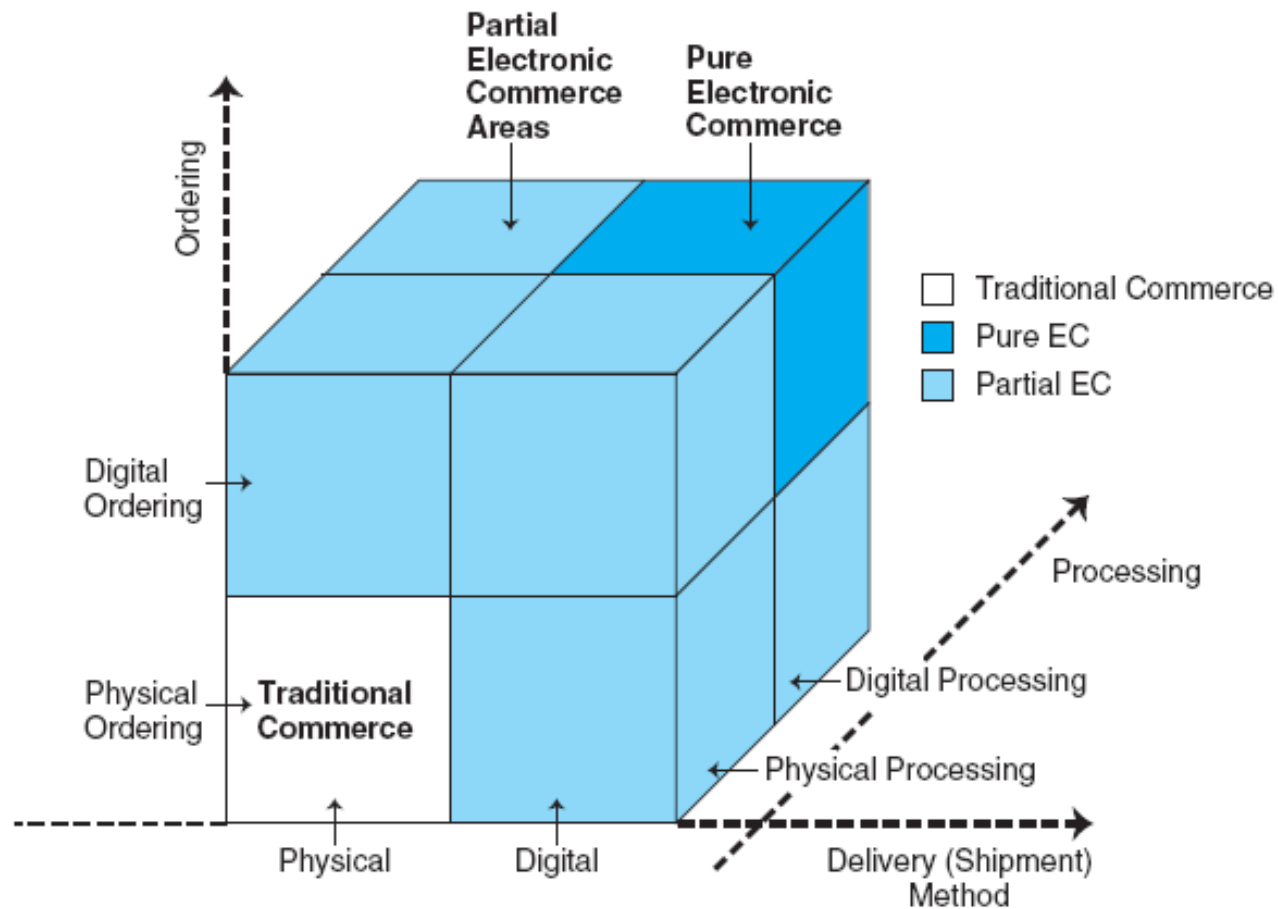
- Organizations that conduct some e-commerce activities, usually as an additional marketing channel

Examples of E-Commerce

- Retailing and Marketing
- Entertainment and Travel
- Education and Learning
- Banking and Investment
- Health-Care
- Auction
- Personal and Business Service

EXHIBIT 1.1

The Dimensions of Electronic Commerce



Electronic Commerce: Definitions and Concepts

- **ELECTRONIC MARKETS AND NETWORKS**

- **electronic market (e-marketplace)**

An online marketplace where buyers and sellers meet to exchange goods, services, money, or information

- **intranet**

An internal corporate or government network that uses Internet tools, such as Web browsers, and Internet protocols

- **extranet**

A network that uses the Internet to link multiple intranets

The Electronic Commerce Field: Classification, Content, and a Brief History

- **AN EC FRAMEWORK**

- EC applications are supported by infrastructure and by the following five support areas:
 1. People
 2. Public policy
 3. Marketing and advertising
 4. Support services
 5. Business partnerships

The Electronic Commerce Field: Classification, Content, and a Brief History

- **CLASSIFICATION OF EC BY THE NATURE OF THE TRANSACTIONS AND THE RELATIONSHIPS AMONG PARTICIPANTS**

- **business-to-business (B2B)**

E-commerce model in which all of the participants are businesses or other organizations

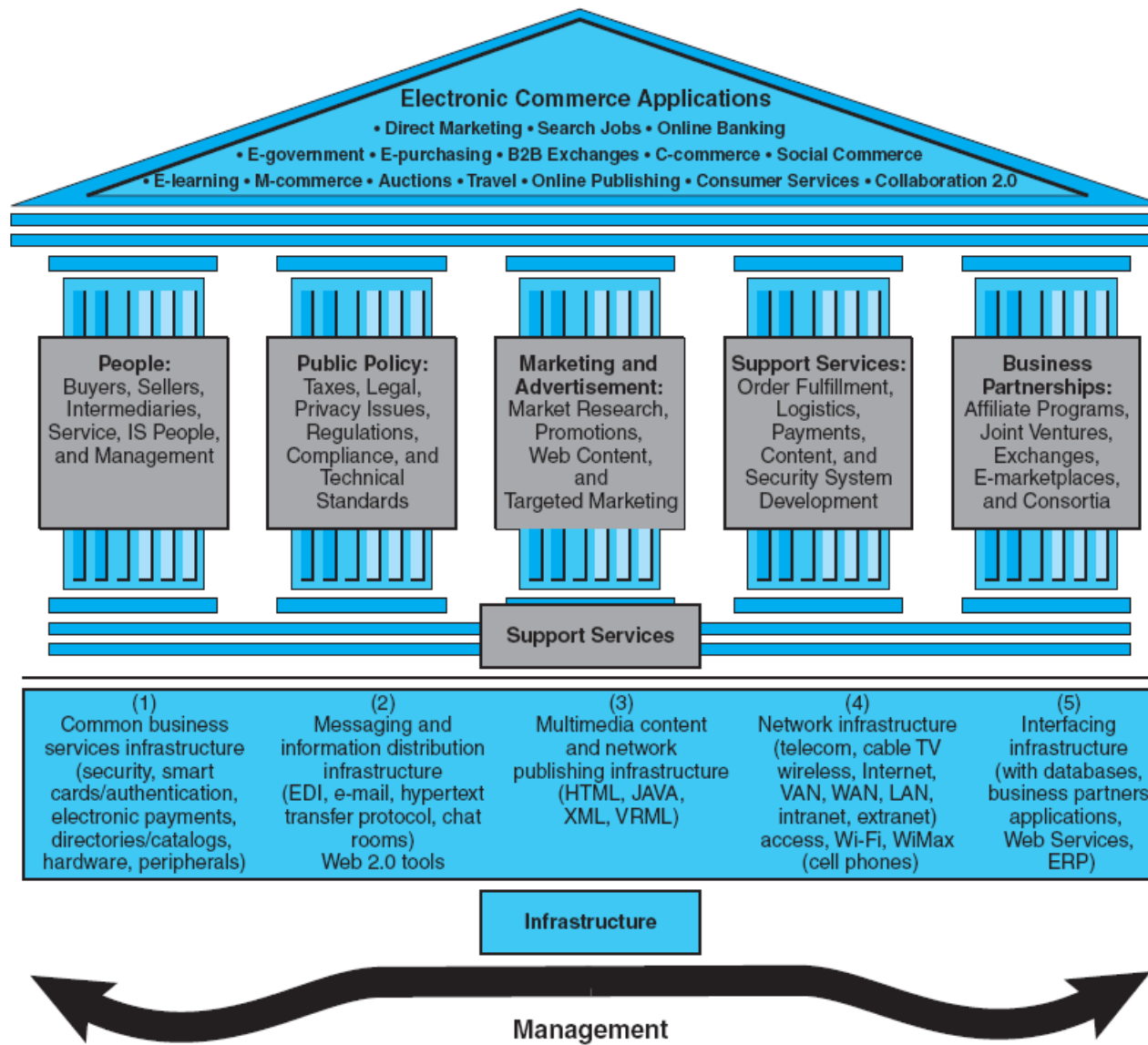
- **business-to-consumer (B2C)**

E-commerce model in which businesses sell to individual shoppers

- **e-tailing**

Online retailing, usually B2C

EXHIBIT 1.3 A Framework for Electronic Commerce



The Electronic Commerce Field: Classification, Content, and a Brief History

- **business-to-business-to-consumer (B2B2C)**
E-commerce model in which a business provides some product or service to a client business that maintains its own customers
- **consumer-to-business (C2B)**
E-commerce model in which individuals use the Internet to sell products or services to organizations or individuals who seek sellers to bid on products or services they need

The Electronic Commerce Field: Classification, Content, and a Brief History

- **intrabusiness EC**

E-commerce category that includes all internal organizational activities that involve the exchange of goods, services, or information among various units and individuals in an organization

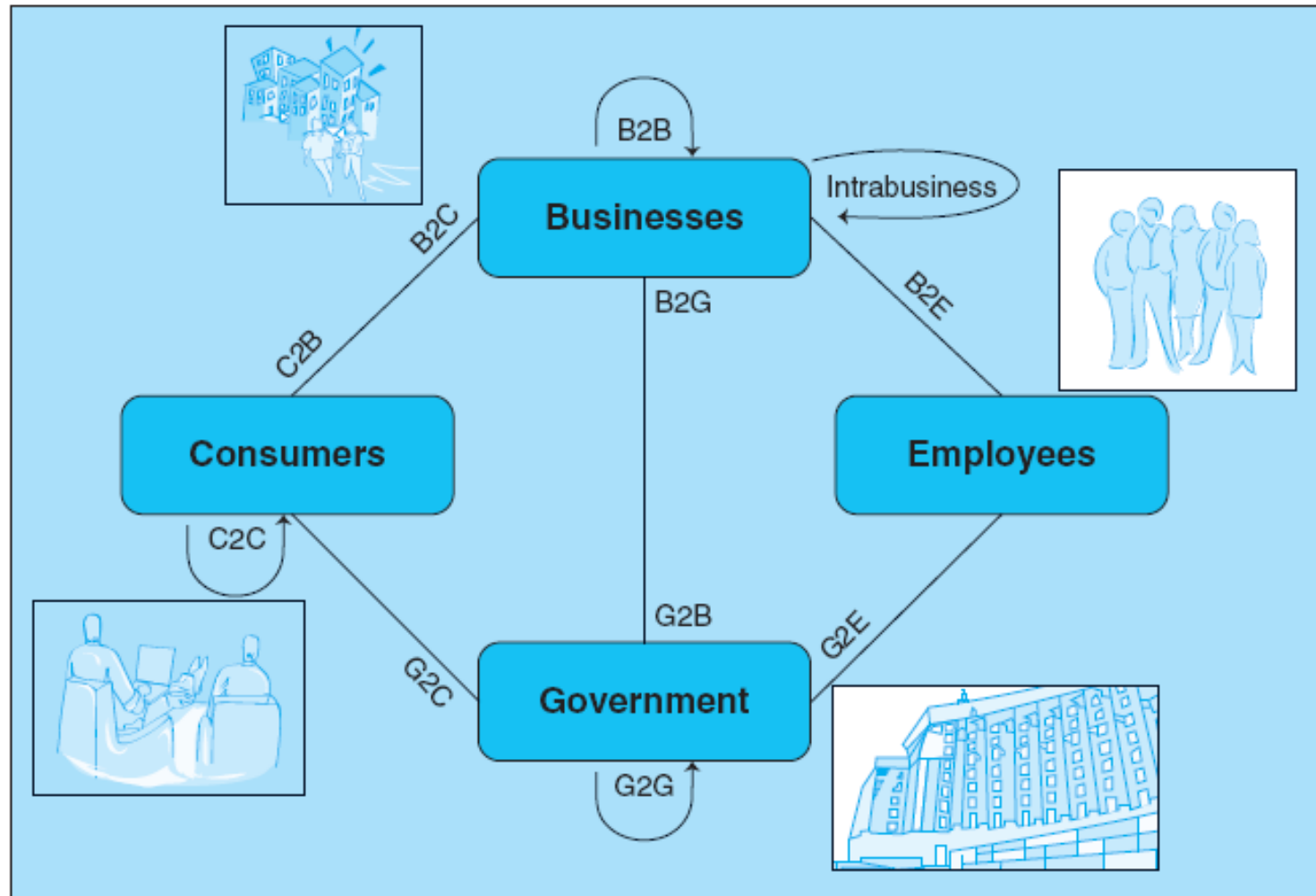
- **business-to-employees (B2E)**

E-commerce model in which an organization delivers services, information, or products to its individual employees

The Electronic Commerce Field: Classification, Content, and a Brief History

- **consumer-to-consumer (C2C)**
E-commerce model in which consumers sell directly to other consumers
- **collaborative commerce (c-commerce)**
E-commerce model in which individuals or groups communicate or collaborate online
- **e-government**
E-commerce model in which a government entity buys or provides goods, services, or information from or to businesses or individual citizens

EXHIBIT 1.4 Categories of Transactions in E-Commerce



The Electronic Commerce Field: Classification, Content, and a Brief History

- **A BRIEF HISTORY OF EC**

- **The Interdisciplinary Nature of EC**
- **The Google Revolution**
- **f-commerce**

E-commerce activities conducted on Facebook or influenced by the site

- **EC Failures**
- **EC Successes**

- **THE FUTURE OF EC**

E-Commerce 2.0:

From Social Commerce to Virtual Worlds

- **social computing**

An approach aimed at making the human–computer interface more natural

- **Web 2.0**

The second generation of Internet-based services that lets people collaborate and share information online in new ways, such as social networking sites, wikis, communication tools, and folksonomies

E-Commerce 2.0: From Social Commerce to Virtual Worlds

- **social network**

A category of Internet applications that help connect friends, business partners, or individuals with specific interests by providing free services such as photo presentation, e-mail, blogging, and so on using a variety of tools

E-Commerce 2.0:

From Social Commerce to Virtual Worlds

- **social networking service (SNS)**

A service that builds online communities by providing an online space for people to build free homepages and that provides basic communication and support tools for conducting different activities in the social network

- **social networking**

The creation or sponsoring of a social network service and any activity, such as blogging, done in a social network (external or internal)

E-Commerce 2.0: From Social Commerce to Virtual Worlds

- **ENTERPRISE SOCIAL NETWORKS**
- **social commerce**

The e-commerce activities conducted in social networks and/or by using social software (i.e., Web 2.0 tools)

E-Commerce 2.0:

From Social Commerce to Virtual Worlds

- **VIRTUAL WORLDS AND SECOND LIFE**
- **virtual world**

A user-defined world in which people can interact, play, and do business; the most publicized virtual world is Second Life

- **How Students Make Money in a Virtual World**
- **THE MAJOR TOOLS OF WEB 2.0**
 - Wikis
 - RSS feeds (**Rich Site Summary**)
 - Blogs
 - Microblogs (e.g., Twitter)

The Digital World: Economy, Enterprises, and Society

- **digital economy**

An economy that is based on digital technologies, including digital communication networks, computers, software, and other related information technologies; also called the *Internet economy*, *the new economy*, or *the Web economy*

The Digital World: Economy, Enterprises, and Society

- **digital enterprise**

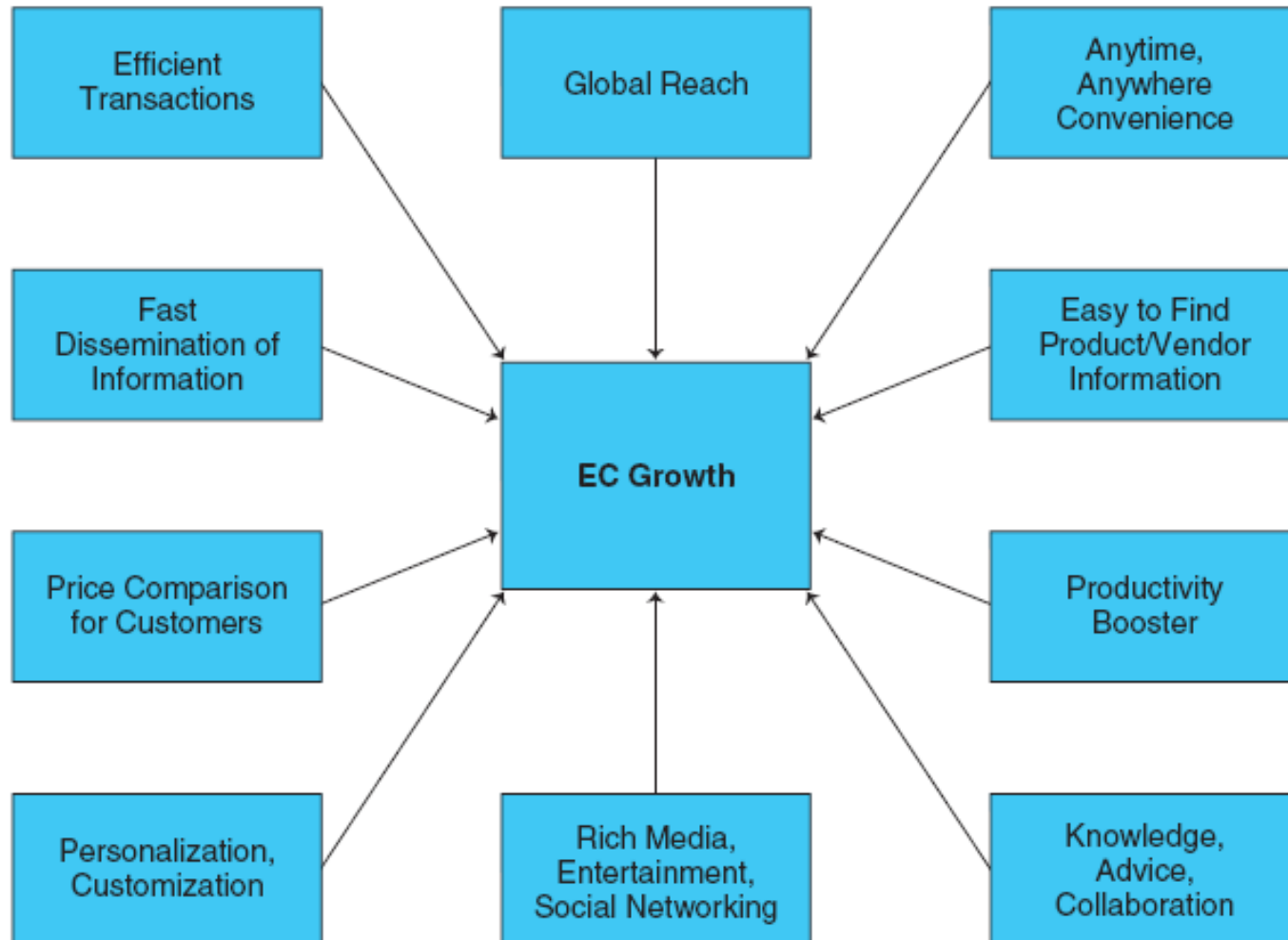
A new business model that uses IT in a fundamental way to accomplish one or more of three basic objectives: reach and engage customers more effectively, boost employee productivity, and improve operating efficiency; uses converged communication and computing technology in a way that improves business processes

- **corporate portal**

A major gateway through which employees, business partners, and the public can enter a corporate website.

- **THE DIGITAL SOCIETY**

EXHIBIT 1.10 Major Capabilities That Contribute to the Growth of EC



Electronic Commerce Business Models

- **business model**

A method of doing business by which a company can generate revenue to sustain itself

Electronic Commerce Business Models

- **THE STRUCTURE AND PROPERTIES OF BUSINESS MODELS**

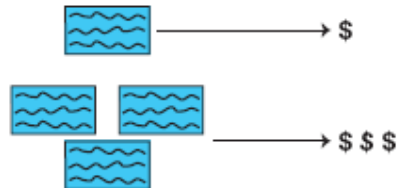
- Revenue Models
- value proposition

The benefits a company can derive from using EC

- **Functions of a Business Model**

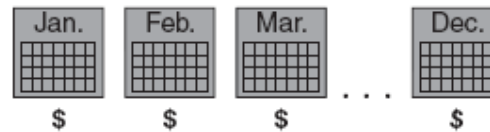
EXHIBIT 1.11 Common Revenue Models

Transaction Fees Model



Commissions paid on volume of transactions

Subscription Model



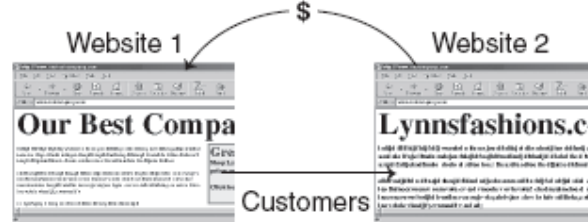
Fixed amounts are charged, usually monthly

Advertisement Model



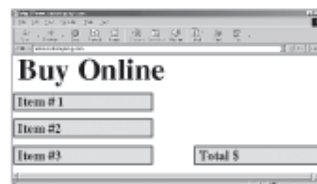
Payments from advertisers

Affiliate Model



Commissions for referring customers

Sales Model



Revenue from sales of goods or services

Electronic Commerce Business Models

- **TYPICAL EC BUSINESS MODELS**

1. **Online direct marketing**
2. **tendering (bidding) system**

Model in which a buyer requests would-be sellers to submit bids; the lowest bidder wins.

3. **Electronic marketplaces and exchanges**
4. **Viral marketing**
5. **Group purchasing**

Managerial Issues

1. Is EC real?
2. Why is B2B e-commerce so essential and successful?
3. Which EC business model should I choose?
4. How can we exploit social commerce?
5. What are the top challenges of EC today?

Summary

1. Definition of EC and description of its various categories
2. The content and framework of EC
3. The major types of EC transactions
4. E-commerce 2.0
5. Description of social commerce and social software

Summary

- 6. The elements of the digital world
- 7. The drivers of EC
- 8. The major EC business models
- 9. Benefits of EC to organizations, consumers, and society
- 10. Limitations of e-commerce